Qrand Quantifier Quiz (Logic week 5)

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Some important basic translations

All P's are Q's	$\forall x(P(x) \rightarrow Q(x))$
Some P's are Q's	$\exists x(P(x) \wedge Q(x))$
No P's are Q's	$\forall x(P(x) \rightarrow \neg Q(x))$
Some P's are not Q's	$\exists x(P(x) \wedge \neg Q(x))$
There is at most one P	$\forall x \forall y((P(x) \wedge P(y)) \rightarrow x = y)$
There is at least one P	$\exists x P(x)$
There is exactly one P	$\exists x \forall y(P(y) \leftrightarrow x = y)$

Question 1

All cubes are small.

A $\forall x\ \text{Small}(x)$

B $\forall x(\text{Cube}(x) \wedge \text{Small}(x))$

C $\forall x(\text{Cube}(x) \rightarrow \text{Small}(x))$

D $\forall x(\text{Small}(x) \rightarrow \text{Cube}(x))$

E $\forall x(\text{Cube}(x) \leftrightarrow \text{Small}(x))$

Question 1

All cubes are small.

A $\forall x\ \text{Small}(x)$

B $\forall x(\text{Cube}(x) \wedge \text{Small}(x))$

C $\forall x(\text{Cube}(x) \rightarrow \text{Small}(x))$

D $\forall x(\text{Small}(x) \rightarrow \text{Cube}(x))$

E $\forall x(\text{Cube}(x) \leftrightarrow \text{Small}(x))$

Only answer C is correct.

Question 2

Some cube is large.

A $\exists x\ \text{Large}(x)$

B $\exists x(\text{Cube}(x) \wedge \text{Large}(x))$

C $\exists x(\text{Cube}(x) \rightarrow \text{Large}(x))$

D $\exists x(\text{Large}(x) \rightarrow \text{Cube}(x))$

E $\exists x(\text{Cube}(x) \leftrightarrow \text{Large}(x))$

Question 2

Some cube is large.

A $\exists x\ \text{Large}(x)$

B $\exists x(\text{Cube}(x) \wedge \text{Large}(x))$

C $\exists x(\text{Cube}(x) \rightarrow \text{Large}(x))$

D $\exists x(\text{Large}(x) \rightarrow \text{Cube}(x))$

E $\exists x(\text{Cube}(x) \leftrightarrow \text{Large}(x))$

Only answer B is correct.

Question 3

No dodecahedron is small.

A $\forall x \neg \text{Dodec}(x) \wedge \forall x \neg \text{Small}(x)$

B $\neg \exists x (\text{Dodec}(x) \wedge \text{Small}(x))$

C $\forall x \neg (\text{Dodec}(x) \rightarrow \text{Small}(x))$

D $\forall x (\text{Dodec}(x) \rightarrow \neg \text{Small}(x))$

E $\exists x \neg (\text{Dodec}(x) \wedge \text{Small}(x))$

F $\neg \exists x \text{Dodec}(x) \wedge \neg \exists x \text{Small}(x)$

Question 3

No dodecahedron is small.

A $\forall x \neg \text{Dodec}(x) \wedge \forall x \neg \text{Small}(x)$

B $\neg \exists x (\text{Dodec}(x) \wedge \text{Small}(x))$

C $\forall x \neg (\text{Dodec}(x) \rightarrow \text{Small}(x))$

D $\forall x (\text{Dodec}(x) \rightarrow \neg \text{Small}(x))$

E $\exists x \neg (\text{Dodec}(x) \wedge \text{Small}(x))$

F $\neg \exists x \text{Dodec}(x) \wedge \neg \exists x \text{Small}(x)$

Both answer B and answer D are correct.

Question 4

There are exactly three cubes.

A $\exists x \exists y \exists z (x \neq y \wedge y \neq z \wedge x \neq z \wedge \forall u[(u = x \vee u = y \vee u = z) \rightarrow \text{Cube}(u)])$

B $\exists x \exists y \exists z (x \neq y \wedge y \neq z \wedge x \neq z \wedge \forall u[(u = x \vee u = y \vee u = z) \leftrightarrow \text{Cube}(u)])$

C $\exists x \exists y \exists z \text{Cube}(x) \wedge \neg \exists u \ \text{Cube}(u)$

D $\exists x \exists y \exists z (x \neq y \wedge y \neq z \wedge x \neq z \wedge \text{Cube}(x) \wedge \text{Cube}(y) \wedge \text{Cube}(z))$

E $\forall x \forall y \forall z \forall u((\text{Cube}(x) \wedge \text{Cube}(y) \wedge \text{Cube}(z) \wedge \text{Cube}(u)) \rightarrow [x = y \vee x = z \vee x = u \vee y = z \vee y = u \vee z = u])$

F $\exists x \exists y \exists z (x \neq y \wedge y \neq z \wedge x \neq z \wedge \text{Cube}(x) \wedge \text{Cube}(y) \wedge \text{Cube}(z) \wedge \forall u[\text{Cube}(u) \rightarrow (u = x \vee u = y \vee u = z)])$

G $\exists x \exists y \exists z (x \neq y \wedge y \neq z \wedge x \neq z \wedge \text{Cube}(x) \wedge \text{Cube}(y) \wedge \text{Cube}(z)) \wedge \forall u(\text{Cube}(u) \rightarrow (u = x \vee u = y \vee u = z))$

H $\text{Cube}(a) \wedge \text{Cube}(b) \wedge \text{Cube}(c) \wedge \neg \exists x (x \neq a \wedge x \neq b \wedge x \neq c \wedge \text{Cube}(x))$

Question 4

There are exactly three cubes.

A
$$\exists x \exists y \exists z (x \neq y \wedge y \neq z \wedge x \neq z \wedge \forall u[(u = x \vee u = y \vee u = z) \rightarrow \text{Cube}(u)])$$

B
$$\exists x \exists y \exists z (x \neq y \wedge y \neq z \wedge x \neq z \wedge \forall u[(u = x \vee u = y \vee u = z) \leftrightarrow \text{Cube}(u)])$$

C $\exists x \exists y \exists z \text{Cube}(x) \wedge \neg \exists u \ \text{Cube}(u)$

D $\exists x \exists y \exists z (x \neq y \wedge y \neq z \wedge x \neq z \wedge \text{Cube}(x) \wedge \text{Cube}(y) \wedge \text{Cube}(z))$

E $\forall x \forall y \forall z \forall u((\text{Cube}(x) \wedge \text{Cube}(y) \wedge \text{Cube}(z) \wedge \text{Cube}(u)) \rightarrow [x = y \vee x = z \vee x = u \vee y = z \vee y = u \vee z = u])$

F $\exists x \exists y \exists z (x \neq y \wedge y \neq z \wedge x \neq z \wedge \text{Cube}(x) \wedge \text{Cube}(y) \wedge \text{Cube}(z) \wedge \forall u[\text{Cube}(u) \rightarrow (u = x \vee u = y \vee u = z)])$

G $\exists x \exists y \exists z (x \neq y \wedge y \neq z \wedge x \neq z \wedge \text{Cube}(x) \wedge \text{Cube}(y) \wedge \text{Cube}(z)) \wedge \forall u(\text{Cube}(u) \rightarrow (u = x \vee u = y \vee u = z))$

H $\text{Cube}(a) \wedge \text{Cube}(b) \wedge \text{Cube}(c) \wedge \neg \exists x (x \neq a \wedge x \neq b \wedge x \neq c \wedge \text{Cube}(x))$

Answers B and F are correct.

Question 4

There are exactly three cubes.

A
$$\exists x \exists y \exists z (x \neq y \wedge y \neq z \wedge x \neq z \wedge \forall u[(u = x \vee u = y \vee u = z) \rightarrow \text{Cube}(u)])$$

B
$$\exists x \exists y \exists z (x \neq y \wedge y \neq z \wedge x \neq z \wedge \forall u[(u = x \vee u = y \vee u = z) \leftrightarrow \text{Cube}(u)])$$

C $\exists x \exists y \exists z \text{Cube}(x) \wedge \neg \exists u \ \text{Cube}(u)$

D $\exists x \exists y \exists z (x \neq y \wedge y \neq z \wedge x \neq z \wedge \text{Cube}(x) \wedge \text{Cube}(y) \wedge \text{Cube}(z))$

E $\forall x \forall y \forall z \forall u((\text{Cube}(x) \wedge \text{Cube}(y) \wedge \text{Cube}(z) \wedge \text{Cube}(u)) \rightarrow [x = y \vee x = z \vee x = u \vee y = z \vee y = u \vee z = u])$

F $\exists x \exists y \exists z (x \neq y \wedge y \neq z \wedge x \neq z \wedge \text{Cube}(x) \wedge \text{Cube}(y) \wedge \text{Cube}(z) \wedge \forall u[\text{Cube}(u) \rightarrow (u = x \vee u = y \vee u = z)])$

G $\exists x \exists y \exists z (x \neq y \wedge y \neq z \wedge x \neq z \wedge \text{Cube}(x) \wedge \text{Cube}(y) \wedge \text{Cube}(z)) \wedge \forall u(\text{Cube}(u) \rightarrow (u = x \vee u = y \vee u = z))$

H $\text{Cube}(a) \wedge \text{Cube}(b) \wedge \text{Cube}(c) \wedge \neg \exists x (x \neq a \wedge x \neq b \wedge x \neq c \wedge \text{Cube}(x))$

Answers B and F are correct. Note that D says "there are at least three cubes" and that E says "there are at most three cubes".